Hidden market regimes in weekly stock-market direction classification: comparing standard and HMM-informed LSTMs

Honours Project Report

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Abstract

This honours project investigates whether Hidden Markov Model (HMM) regime information improves Long Short-Term Memory (LSTM) forecasts of next-week stock-price direction for Apple Inc. (AAPL), International Business Machines Corporation (IBM), and Microsoft Corporation (MSFT). The project initially considered daily next-day return regression, but those models produced low-variance forecasts and near-baseline directional behaviour; the final task was therefore reformulated as weekly binary classification. A two-state Gaussian HMM was fitted separately for each ticker to construct causal regime probabilities, and the primary analysis compared technical-only LSTMs with otherwise identical technical-plus-HMM LSTMs under chronological data splits, training-only scaling, validation-based lookback and threshold selection, and five matched random seeds. The comparison was supported by HMM-only, technical logistic-regression, training-majority, and persistence benchmarks, while paired moving-block bootstrap intervals and a nine-fold rolling replication for AAPL and IBM were used to assess robustness.

The HMMs produced economically interpretable bullish/bearish or calm/stress regimes with persistent transition structures, but their predictive contribution was substantially weaker. In the controlled test, adding the HMM probability reduced AAPL LSTM ROC-AUC from 0.531 to 0.470, produced only a negligible IBM increase from 0.583 to 0.585 while lowering balanced accuracy, and left MSFT essentially unchanged below random ranking. Technical logistic regression achieved the highest balanced accuracy for AAPL and IBM, whereas persistence performed best for MSFT. The bootstrap results confirmed deterioration for AAPL but did not establish a non-zero HMM effect for IBM or MSFT, and the completed rolling replication also failed to show a stable incremental benefit across market periods. Broader exploratory market-context models achieved stronger best-observed ROC-AUC values, although these results were selected retrospectively and remain hypothesis-generating. Overall, HMMs described latent market conditions more consistently than they improved next-week directional prediction, so the evidence does not support a reliable forecasting advantage from adding an isolated HMM-derived probability to the technical LSTM.

Contents

1 Introduction

The predictability of financial markets has long been one of the central debates in financial economics. According to the Efficient Market Hypothesis, asset prices should reflect available information, making it difficult to consistently forecast future price movements using publicly available information alone [1]. This view is closely connected to the random-walk interpretation of stock prices, where short-term movements are difficult to infer from past prices. Nevertheless, forecasting remains practically important: investors, risk managers, and researchers are often less interested in predicting exact prices than in identifying whether market conditions are likely to become more favorable or unfavorable. This motivates a shift from return regression to market-state classification. Instead of forecasting the exact next-period return, the objective becomes predicting whether the next period is more likely to be bullish or bearish. This is closely related to the literature on bull and bear market identification, where the equity-market state is treated as a latent variable. Kole and van Dijk [2] show that regime-switching models are particularly relevant for out-of-sample forecasting because they incorporate both return means and volatility, whereas rule-based methods are more suitable for in-sample identification of past market states.

Recent developments in machine learning have renewed interest in this problem. Long Short-Term Memory neural networks are designed for sequential data and can capture temporal dependencies through memory mechanisms [3]. Hidden Markov Models provide a complementary framework by modelling observed data as being generated by unobserved regimes [4]. Combining these two approaches is therefore attractive: the LSTM can learn nonlinear temporal patterns, while the HMM can provide regime-based information about the market environment.

Previous studies have explored similar hybrid approaches in economic forecasting. Zhang et al. [5] apply an LSTM-HMM framework to China's GDP fluctuation-state prediction and compare HMM, GMM-HMM, and LSTM-HMM models across different rolling time windows and input frequencies. Sivakumar [6] also applies an HMM-LSTM fusion model, where HMM-derived hidden states and state means are added as LSTM features, and uses Integrated Gradients to improve interpretability. These studies suggest that HMM-LSTM models can be useful, but they also raise an important question: does regime information genuinely improve out-of-sample prediction, or does it mainly provide an interpretable description of past states?

This question is especially important for stock prediction. Daily stock returns are noisy and weakly predictable, so strong in-sample fit can easily overstate true forecasting ability. Complex models such as LSTMs may collapse toward predicting the average return or majority direction when the signal-to-noise ratio is low. Moreover, HMM states labelled as bullish or bearish are not always literal positive- and negative-return states; they may instead separate calm and volatile periods. Therefore, it is necessary to test HMM-derived regime probabilities in an out-of-sample setting rather than assuming that they automatically improve prediction.

The project initially considered daily next-day return regression before moving to weekly next-week direction classification. This change should not be interpreted as assuming that weekly data are intrinsically less noisy than daily data. Weekly aggregation changes the forecasting horizon and the scale of the target, while also reducing the number of available observations. The suitability of the weekly formulation must therefore be assessed empirically through out-of-sample performance rather than assumed in advance. Accordingly, the primary research question is:

> *To what extent does HMM-derived regime information improve LSTM-based prediction of next-week stock direction beyond stock-specific technical features for Apple Inc. (AAPL), International Business Machines Corporation (IBM), and Microsoft Corporation (MSFT)?*

For stock i in week w, the weekly log return is defined as $r_{i,w} = \log(P_{i,w}) - \log(P_{i,w-1})$, where $P_{i,w}$ denotes the weekly closing price. The next-week directional target is

$$Z_{i,w+1} = \begin{cases} 1, & r_{i,w+1} > 0, \\ 0, & r_{i,w+1} \leq 0. \end{cases}$$

Thus, information available through week w is used to predict whether the return in week $w+1$ is positive or non-positive.

The primary empirical analysis compares a technical-only LSTM with an otherwise identical technical-plus-HMM LSTM. The comparison is supported by a training-majority baseline, a persistence baseline, an HMM-only probability benchmark, and technical logistic regression. Additional experiments involving the Invesco QQQ exchange-traded fund (QQQ), the Cboe Volatility Index (VIX), the Cboe Nasdaq-100 Volatility Index (VXN), and stock-relative market features are treated as secondary analyses rather than as part of the primary controlled comparison.

2 Literature Review

The Efficient Market Hypothesis argues that if markets rapidly incorporate available information, consistent abnormal return prediction should be difficult [1]. This does not imply that all forms of prediction are impossible, but it means that empirical forecasting claims must be evaluated carefully, especially in noisy financial time series. For this reason, recent stock-forecasting studies often formulate the problem as directional classification rather than exact return regression. For example, Borovkova and Tsiamas [7] study high-frequency stock-market classification using an ensemble of LSTM networks and evaluate performance using the area under the receiver operating characteristic curve (ROC-AUC). ROC-AUC is a threshold-independent ranking measure that represents the probability that a randomly chosen positive observation receives a higher predicted score than a randomly chosen negative observation; a value of 0.50 corresponds to random ranking. Their work motivates two important design choices in this report: formulating stock prediction as a direction-classification problem and evaluating models using time-aware out-of-sample metrics rather than relying only on in-sample fit.

Long Short-Term Memory networks were introduced by Hochreiter and Schmidhuber [3] to address the difficulty of learning long-term dependencies in recurrent neural networks. Their gated structure allows the model to selectively retain, update, and output information from sequential inputs, making LSTMs suitable for financial time series where returns, volatility, momentum, and drawdown evolve over time. Hidden Markov Models provide a complementary perspective by assuming that observations are generated by unobserved regimes [4]. In financial applications, such regimes can be interpreted as latent market states, although they may capture volatility differences as much as bullish or bearish return behaviour. This is consistent with Kole and van Dijk [2], who show that regime-switching models are useful for out-of-sample bull and bear market forecasting because they incorporate both return means and volatility.

Hybrid HMM-LSTM models attempt to combine these strengths. The HMM provides compact regime information, while the LSTM uses sequential features to form nonlinear predictions. Zhang et al. [5] apply an LSTM-HMM framework to China's GDP fluctuation-state prediction and compare different input frequencies and rolling windows, finding that the LSTM-HMM generally performs better than benchmark HMM-based alternatives. Similarly, Sivakumar [6] adds HMM-derived hidden states and state means as LSTM features for economic forecasting and uses Integrated Gradients, an attribution method introduced by Sundararajan et al. [8], to examine the contribution of individual model inputs. Building on these ideas, the present report tests whether HMM-derived regime probabilities and weekly market-context variables improve LSTM-based next-week stock-direction prediction for AAPL, IBM, and MSFT. An Integrated Gradients analysis was initially planned as an additional interpretability exercise, but it was not completed within the scope of the project and is therefore left for future work.

3 Data and Feature Engineering

This section describes the construction of the weekly modelling dataset used in the empirical analysis. The project initially considered next-day return regression using daily observations, but the final framework predicts the direction of the following weekly return. The data-processing pipeline therefore consists of four main stages: aggregating the daily observations to weekly frequency, defining the next-week classification target, constructing stock-specific and market-context predictors, and preparing leakage-aware sequential inputs for the models.

3.1 Data Aggregation and Prediction Target

The empirical analysis uses daily open, high, low, close, and volume (OHLCV) observations obtained from Yahoo Finance for Apple Inc. (AAPL), International Business

Machines Corporation (IBM), and Microsoft Corporation (MSFT). Each daily observation contains

$$\{\text{Ticker}, \text{Date}, \text{Open}, \text{High}, \text{Low}, \text{Close}, \text{Volume}\}.$$

The three stocks were selected for three main reasons. First, they are large and highly liquid equities with long and comparatively complete historical price series. High liquidity reduces the influence of infrequent trading and makes the observed prices more representative of current market conditions. Second, all three firms are technology-related companies and are therefore exposed to some common sector-wide and macroeconomic shocks. This provides a controlled comparative setting in which to examine whether similar market stress periods produce similar latent regimes. Third, the stocks nevertheless differ substantially in their long-run price trajectories and volatility behaviour, allowing the analysis to test whether the same modelling framework generalises across different firms.

The common sector classification is used to motivate comparison rather than to pool the three stocks into one time series. Each HMM and LSTM is estimated separately for each ticker. Consequently, the models do not directly use the other two stocks as predictor variables; instead, the shared sector exposure provides a common economic context for comparing the inferred regimes.

To compare the long-run trajectories on a common scale, each weekly closing-price series is indexed to 100 at the first common model week. Figure 1 also highlights the COVID-19 crash and the 2022 inflation and monetary-tightening bear market. These periods provide descriptive reference points for evaluating whether the HMM assigns elevated bearish probabilities during recognised episodes of market stress.

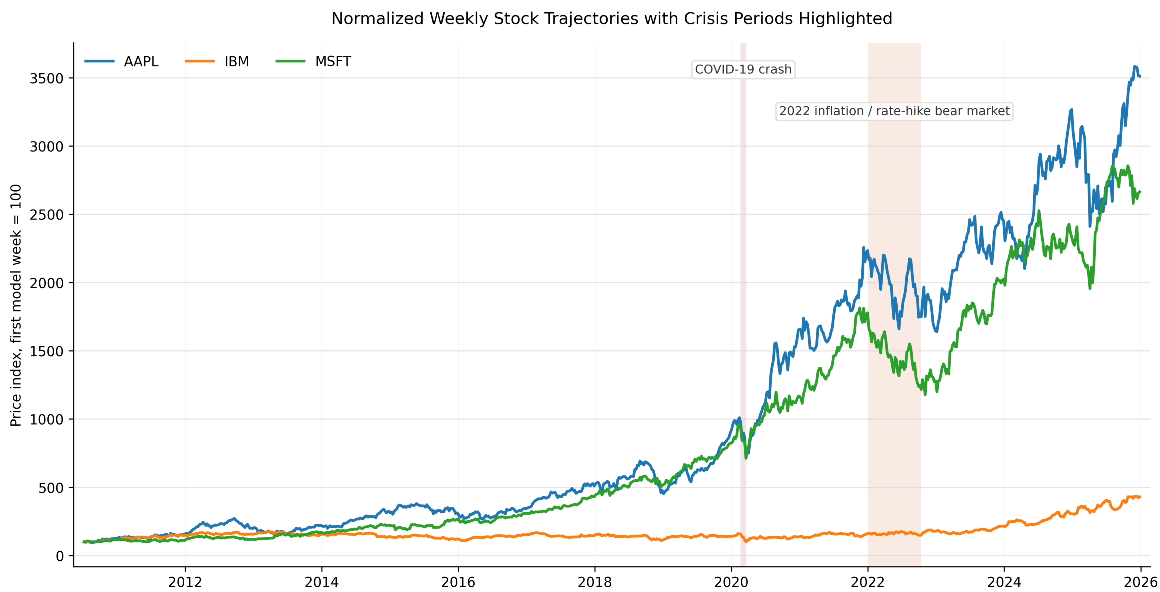


Figure 1: Indexed weekly closing-price trajectories for AAPL, IBM, and MSFT. Each series is normalised to 100 at the first common model week. The shaded regions mark the COVID-19 crash and the 2022 inflation and rate-hike bear market.

Figure 1 shows substantial differences in the long-run trajectories of the three stocks. AAPL and MSFT experienced considerably stronger cumulative growth than IBM, but they also displayed more pronounced declines during the highlighted market-stress periods. IBM was comparatively stable, particularly during the 2022 inflation and rate-hike bear market, when its indexed price declined less sharply than those of AAPL and MSFT. However, this relative resilience was accompanied by substantially weaker long-run price appreciation. The figure therefore illustrates a descriptive trade-off between growth and stability: AAPL and MSFT achieved greater cumulative gains but experienced larger drawdowns, whereas IBM combined lower growth with comparatively smaller crisis-related fluctuations.

Despite these differences, the three stocks were exposed to the same broad market shocks. This combination of common sector-wide conditions and ticker-specific behaviour motivates estimating the models separately while comparing their inferred regimes across the same historical episodes. These observations are descriptive rather than a formal risk-adjusted performance comparison; volatility and drawdown are examined more directly through the engineered features and HMM diagnostics.

The original version of the project used daily observations and next-day log return as the prediction target. For stock i on trading day d, the daily return was defined as $r_{i,d} = \log(P_{i,d}) - \log(P_{i,d-1})$, where $P_{i,d}$ denotes the daily closing price. Although the daily formulation provided a larger number of observations, it created a weak forecasting problem. Daily stock returns are generally small, noisy, and concentrated around zero. Consequently, the early LSTM models could obtain relatively low regression losses by producing predictions close to the average return while providing little useful directional discrimination.

Figure 2 illustrates this difficulty: the 60-day rolling mean returns fluctuate around zero, and their confidence bands overlap substantially across all three stocks and time periods. Moreover, the daily experiments also produced low-variance predictions and, in some cases, a strong tendency to predict the positive class. These results motivated treating the daily analysis as a diagnostic stage and reformulating the final task at the weekly frequency.

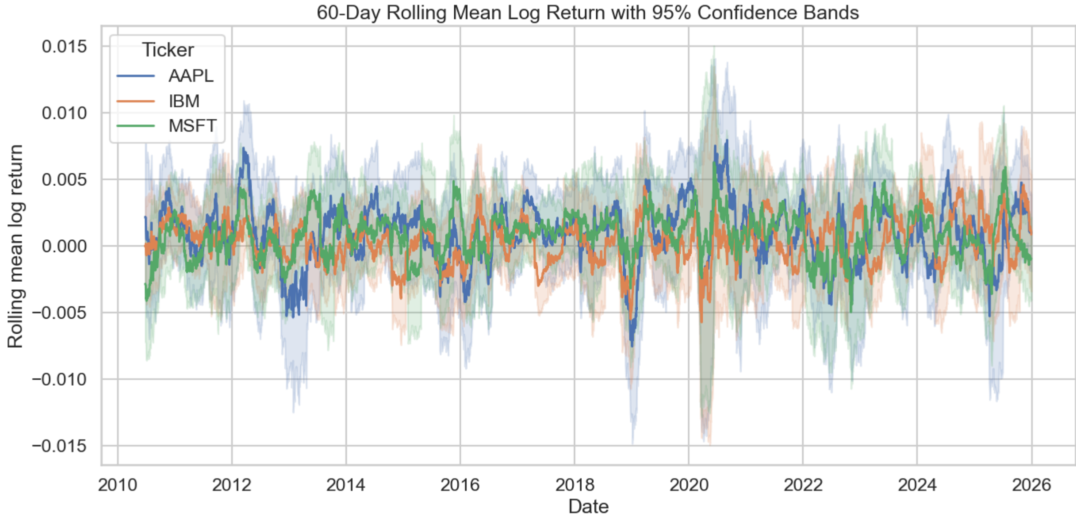


Figure 2: Daily 60-day rolling mean log returns with confidence bands.

Daily observations are aggregated into Friday-ending weeks. For stock i and week w, the weekly OHLCV variables are constructed as

$$\begin{aligned}
\text{Open}_{i,w} &= \text{first observed opening price in week } w, \\
\text{High}_{i,w} &= \max(\text{daily high prices in week } w), \\
\text{Low}_{i,w} &= \min(\text{daily low prices in week } w), \\
\text{Close}_{i,w} &= \text{last observed closing price in week } w, \\
\text{Volume}_{i,w} &= \sum \text{daily trading volume in week } w.
\end{aligned}$$

This aggregation preserves the conventional financial interpretation of weekly OHLCV data. The weekly open and close represent the beginning and end of the trading week, the high and low describe the price range within the week, and the weekly volume measures total trading activity.

For each stock i, the weekly log return is defined as $r_{i,w} = \log(P_{i,w}) - \log(P_{i,w-1})$, where $P_{i,w}$ denotes the weekly closing price. The supervised target is the direction of the following weekly return:

$$Z_{i,w+1} = \begin{cases} 1, & r_{i,w+1} > 0, \\ 0, & r_{i,w+1} \leq 0. \end{cases}$$

A target value of 1 is interpreted as a bullish week, while a value of 0 is interpreted as a bearish week. Each feature vector observed at week w is therefore paired with the realized direction in week $w + 1$. This alignment ensures that only information available through week w is used to predict the subsequent return direction.

The supervised direction label should not be confused with the latent HMM regime labels introduced later. The supervised target is determined directly by the sign of the realized next-week return. In contrast, an HMM regime is an estimated latent state and may distinguish calm and stressed conditions rather than strictly positive and negative return periods.

3.2 Technical and Market-Context Features

The baseline LSTM does not use the weekly OHLCV variables directly. Instead, it receives 15 engineered stock-specific predictors designed to summarize return, trading activity, volatility, momentum, trend, and drawdown. Using transformed variables also prevents large differences in raw price or volume scale from dominating the input. Table 1 summarizes the technical feature set: the 4-, 12-, and 26-week windows represent approximately one month, three months, and six months of market history.

Table 1: Stock-specific weekly technical features used in the baseline LSTM.

Feature group	**Variables**	**Definition and interpretation**
Return	Weekly close-to-close log return; weekly open-to-close log return	Measures the stock's close-to-close return and its price movement within the trading week.
Range	Weekly high-low range	Calculated as $(\text{High}_{i,w} - \text{Low}_{i,w})/\text{Close}_{i,w}$; captures within-week price dispersion.
Volume	Weekly log-volume change	Measures changes in trading activity without allowing differences in raw volume scale to dominate the model.
Volatility	Rolling volatility over 4, 12, and 26 weeks	Standard deviation of weekly returns over short- and medium-term windows.
Momentum	Cumulative return over 4, 12, and 26 weeks	Measures the direction and magnitude of recent accumulated performance.
Trend	Simple moving-average gaps over 4, 12, and 26 weeks	For $h \in \{4, 12, 26\}$, calculated as $\text{MA\_Gap}_{i,w}^{(h)} = P_{i,w}/\overline{P}_{i,w}^{(h)} - 1$. Positive values indicate that the stock trades above its recent trend, while negative values indicate that it trades below its recent trend.
Drawdown	Drawdown over 12 and 26 weeks	Calculated relative to the maximum closing price within the corresponding window; measures decline from a recent peak.

More precisely, the trailing simple moving average is defined as $\overline{P}_{i,w}^{(h)} = \frac{1}{h}\sum_{j=0}^{h-1} P_{i,w-j}$, where $P_{i,w}$ denotes the adjusted weekly closing price of stock i in week w and $h \in \{4, 12, 26\}$.

The technical-only feature set is subsequently extended with broader market and volatility information. The Invesco QQQ exchange-traded fund (QQQ), which tracks the Nasdaq-100 Index, is used as a technology-heavy market proxy. The Cboe Volatility Index (VIX) measures market expectations of near-term volatility implied by S&P 500 index options, whereas the Cboe Nasdaq-100 Volatility Index (VXN) provides the corresponding measure based on Nasdaq-100 index options. QQQ therefore represents common market-price movements, while VIX and VXN represent broader-market and Nasdaq-focused volatility conditions. These variables are included because the weekly direction of an individual stock may reflect both firm-specific behaviour and common market-wide risk conditions.

All external variables are aligned to the same Friday-ending weekly frequency as the individual-stock observations. Compact context specifications use a restricted number of transformations to limit the risk of overfitting. Fuller specifications additionally include changes, returns, rolling measures, and stock-versus-market comparisons.

One conceptually important context variable is the stock's return relative to QQQ:

$$r_{i,w}^{\text{rel}} = r_{i,w} - r_{\text{QQQ},w}.$$

A positive relative return indicates that the stock outperformed QQQ during week w, whereas a negative value indicates underperformance. This feature separates stock-specific performance from movements that were shared with the broader technology market.

The technical and context variables are kept in distinguishable feature groups throughout the experiments. This makes it possible to compare whether prediction is primarily supported by the stock's own recent history, broader market conditions, or the additional HMM regime information.

3.3 HMM-Derived Regime Features

HMM-derived variables are constructed only after the weekly return series has been prepared and the HMM has been estimated. They are therefore treated as model-generated regime features rather than ordinary transformations of the raw OHLCV data.

For stock i, let $S_{i,w}$ denote the latent HMM state in week w, and let $\mathcal{F}_{i,w}$ denote the information available through that week. The HMM produces a probability distribution over the current latent state and uses the estimated transition matrix to obtain next-week probabilities. The principal regime feature is therefore written as $P\left(S_{i,w+1} = \text{Bullish} \mid \mathcal{F}_{i,w}\right).$

The experiments also use an HMM-implied next-week positive-return probability, $P\left(Z_{i,w+1} = 1 \mid \mathcal{F}_{i,w}\right)$, which combines the predicted state probabilities with the training-sample positive-return frequency associated with each state.

The state numbers estimated by an HMM have no intrinsic economic interpretation. State labels are consequently assigned using training data only, with the higher-return state labelled bullish and the lower-return state labelled bearish. Nevertheless, the labels remain relative. A state called bearish may still have a positive mean return if its return is lower, or its volatility higher, than that of the alternative state.

The HMM probabilities are therefore interpreted as summaries of latent market conditions rather than direct observations of the true next-week direction. Keeping this distinction explicit is important because the central empirical question is whether these regime summaries provide incremental information beyond the technical features already supplied to the LSTM.

3.4 Chronological Splitting, Scaling, and Sequence Construction

After weekly aggregation, feature construction, rolling-window warm-up, and removal of observations without an available next-week target, the final dataset contains 808 usable observations for each ticker. The observations are divided chronologically into training, validation, and test samples according to the target date rather than the feature date. This distinction is necessary because the label attached to the feature row in week w belongs to week $w + 1$.

Table 2: Final weekly dataset split by ticker.

Ticker	Train rows	Validation rows	Test rows	Total rows
AAPL	565	121	122	808
IBM	565	121	122	808
MSFT	565	121	122	808
Total	1695	363	366	2424

The test sample covers September 1, 2023, through December 26, 2025. Training observations are used to estimate model parameters, feature scalers, HMM parameters, and other training-dependent quantities. The validation sample is used for choices such as lookback length and classification threshold. The test sample is reserved for final out-of-sample evaluation.

All scaling is performed separately for each ticker and without look-ahead bias. Each scaler is fitted using training observations only and is then applied unchanged to the corresponding training, validation, and test observations. Validation and test information therefore cannot influence either the location or scale of the training features.

The LSTM receives sequences of weekly feature vectors. For lookback length L, the sequence used to predict stock i at week $w + 1$ is

$$X_{i,w}^{(L)} = \left(x_{i,w-L+1}, x_{i,w-L+2}, \dots, x_{i,w}\right),$$

where $x_{i,w}$ denotes the feature vector observed in week w. The corresponding label is $Z_{i,w+1}$. The resulting LSTM input tensor has dimensions

$$(n, L, p),$$

where n is the number of sequences, L is the number of historical weeks, and p is the number of input features.

The candidate lookback lengths are $L \in \{4, 8, 12, 26\}$, corresponding approximately to one month, two months, three months, and six months of market history. However, exploratory experiments considered lookbacks of $L \in \{4, 8, 12, 26\}$, whereas the final controlled comparison used the restricted validation grid $L \in \{4, 8, 12\}$. These alternatives test whether the next-week direction is better explained by short-term patterns or by a longer sequence of market information. To be more specific, a sequence is assigned to the training, validation, or test sample according to the split of its target observation. A validation or test sequence may include earlier feature observations from the preceding split because these observations would already be known at the prediction date. The target itself, however, always belongs to the relevant validation or test period. This construction maintains chronological realism while avoiding the unnecessary removal of the first $L - 1$ target observations from each new split.

Overall, the resulting dataset provides a leakage-aware basis for comparing technical-only, context-aware, and HMM-informed sequence models. The same observed target, chronological split, and evaluation samples are maintained across model families so that any performance difference can be attributed to the additional information or modelling structure rather than to changes in the underlying dataset.

4 Methodology

4.1 Research design and primary controlled comparison

The primary research question is whether an HMM-derived probability provides incremental out-of-sample value for weekly LSTM stock-direction forecasting beyond stock-specific technical features. This question is evaluated through a controlled feature-comparison design. For each ticker, a technical-only LSTM is compared with a technical-plus-HMM LSTM. At each candidate lookback, the models use the same observations, target, technical predictors, chronological split, neural-network architecture, training procedure, random seeds, and validation rules; the only input difference is the addition of one HMM-implied next-week up-probability feature. However, the lookback is selected separately for each feature configuration using the same validation procedure, so the final selected pipelines may use different lookbacks. At each candidate lookback, the paired models differ only by the HMM-derived feature. Because lookback is subsequently selected separately for each feature configuration, however, the final reported differences estimate the effect of adding the HMM feature within the complete validation-selected pipeline rather than a strictly fixed-lookback feature effect.

As described in Section 3, the analysis uses ticker-specific weekly observations for AAPL, IBM, and MSFT. The binary target is

$$y_{i,w+1} = \mathbf{1}\{r_{i,w+1} > 0\},$$

where $r_{i,w+1}$ denotes the following weekly log return for stock i. A zero return is assigned to the non-positive class. Observations are divided chronologically by target date into 565 training rows, 121 validation rows, and 122 test rows per ticker before sequence construction. Validation observations are used for model and threshold selection, whereas the test observations are reserved for final out-of-sample evaluation.

The comparison between the two LSTM feature sets is treated as the primary controlled analysis. Experiments involving volatility context, QQQ, market-relative variables, alternative HMM specifications, calibration diagnostics, and rolling evaluation are treated as secondary or robustness analyses.

4.2 HMM specification and predictive feature construction

For the primary fixed-split analysis, a separate two-state Gaussian Hidden Markov Model is estimated for each ticker. The observed variable is the weekly log return, standardized using a ticker-specific `StandardScaler` fitted on training observations only.

Let $S_{i,w}$ denote the latent state of stock i in week w. The hidden-state process is assumed to follow a first-order Markov chain:

$$S_{i,w} \in \{0,1\}, \qquad \Pr(S_{i,w} = b \mid S_{i,w-1} = a) = A_{i,ab},$$

where A_i is the ticker-specific transition matrix and $A_{i,ab}$ denotes the probability of moving from state a in week $w-1$ to state b in week w.

Let $\widetilde{r}_{i,w}$ denote the standardized weekly return, defined as

$$\widetilde{r}_{i,w} = \frac{r_{i,w} - \overline{r}_{i,\text{train}}}{s_{i,\text{train}}},$$

where $\overline{r}_{i,\text{train}}$ and $s_{i,\text{train}}$ are, respectively, the mean and scale of the weekly returns in the training sample for ticker i, as estimated by the training-fitted `StandardScaler`. Conditional on the latent state, the standardized weekly return follows

$$\widetilde{r}_{i,w} \mid S_{i,w} = k \sim \mathcal{N}(\mu_{i,k}, \sigma^2_{i,k}).$$

The implemented specification uses `GaussianHMM` with two components and a full covariance specification, a maximum of 500 iterations, tolerance 10^{-4}, a minimum covariance of 10^{-4}, and random seed 42. Although a full covariance specification is used, the primary emission distribution is univariate because only `Weekly_Log_Return` enters the HMM.

The initial-state probabilities, transition matrix A_i, and Gaussian emission parameters were estimated jointly by maximum likelihood using the expectation–maximization algorithm implemented by `GaussianHMM`. All HMM parameters were estimated using training observations only.

The numerical state identifiers have no intrinsic economic meaning. After estimation, states are labelled using training observations only. The state with the higher training-sample mean weekly return is labelled bullish, and the other state is labelled bearish. These labels are relative: a state labelled bearish may still contain some positive-return weeks and may also differ from the bullish state in volatility.

The predictive HMM feature is based on causal filtered probabilities rather than full-sample smoothed probabilities. For each ticker, the forward recursion produces

$$\alpha_{i,w,k} = \Pr(S_{i,w} = k \mid \widetilde{r}_{i,1:w}),$$

which conditions only on observations available through week w. The one-step-ahead state probabilities are then calculated as

$$\pi^{S}_{i,w+1,k} = \sum_{a=0}^{1} \alpha_{i,w,a} A_{i,ak}.$$

Let $\widehat{S}_{i,w}$ denote the hard HMM state assignment for a training observation, and let $\mathcal{T}_i$ denote the training sample for ticker i. The state-specific positive-return frequency is estimated as

$$\widehat{u}_{i,k} = \frac{\sum_{w \in \mathcal{T}_i} \mathbf{1}\{\widehat{S}_{i,w} = k\}\mathbf{1}\{r_{i,w} > 0\}}{\sum_{w \in \mathcal{T}_i} \mathbf{1}\{\widehat{S}_{i,w} = k\}}.$$

The HMM-implied next-week up probability is then

$$q_{i,w+1} = \sum_{k=0}^{1} \pi^{S}_{i,w+1,k} \widehat{u}_{i,k}.$$

This quantity was saved in the upstream HMM pipeline as `HMM_Positive_Return_Prob_Next`. In the controlled comparison, it was renamed to `HMM_Up_Prob_Raw`. The HMM-only benchmark uses this raw probability, whereas the HMM-augmented LSTM uses `HMM_Up_Prob_Scaled`, obtained from a separate ticker-specific `StandardScaler` fitted on training observations only.

Only this single HMM-derived feature is added to the LSTM. The complementary down probability is omitted because it is mechanically determined by the up probability in the binary setting. The primary controlled LSTM does not use the raw HMM state, `HMM_Bullish_Prob_Next`, smoothed state probabilities, or future return information.

4.3 LSTM Classifiers and Feature Ablation

The technical-only LSTM uses the 15 stock-specific weekly predictors defined in Table 1. These variables summarize weekly return, intraweek price movement, price range, volume change, volatility, momentum, trend, and drawdown. All technical variables are standardized separately by ticker using training observations only.

For a candidate lookback L, the sequence available at week w is

$$X_{i,w}^{(L)} = (x_{i,w-L+1}, x_{i,w-L+2}, \ldots, x_{i,w}) ,$$

where $x_{i,w}$ denotes the feature vector observed in week w. The corresponding target is $y_{i,w+1}$. For either feature configuration, the LSTM maps the historical sequence to an estimated next-week up probability:

$$\widehat{p}_{i,w+1} = f_\theta\left(X_{i,w}^{(L)}\right) ,$$

where f_θ denotes the LSTM classifier and θ denotes its trainable parameters.

The technical-only model has $p = 15$ input variables. The technical-plus-HMM model has $p = 16$, consisting of the same 15 technical variables plus `HMM_Up_Prob_Scaled`. No QQQ, VIX, VXN, recession, market-relative, cross-stock, target, future-return, or smoothed-HMM variables enter the primary controlled comparison.

The classifier is trained using binary cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{n} \sum_{j=1}^{n} [y_j \log(\widehat{p}_j) + (1 - y_j) \log(1 - \widehat{p}_j)] .$$

Table 3 summarizes the architecture and training settings used in the controlled comparison.

Table 3: Final controlled LSTM architecture and training settings.

Component	**Controlled setting**
Input shape	(L, p), where $L \in \{4, 8, 12\}$ and $p = 15$ or 16
LSTM layers	One LSTM layer
Hidden units	16
Dense layers	One dense layer with 8 ReLU units
Dropout	0.20 in the LSTM layer and after the dense layer
Recurrent dropout	0.00
Output layer	One sigmoid unit interpreted as $\Pr(y_{i,w+1} = 1)$
Loss function	Binary cross-entropy
Optimizer	Adam
Learning rate	0.001
Batch size	32
Maximum epochs	150
Early stopping	Validation ROC-AUC, patience 12, with restoration of the best weights
Learning-rate adjustment	Reduction of the learning rate when validation loss reached a plateau
Random seeds	7, 19, 42, 91, and 123
Class weights	None

4.4 Training, Model Selection, and Benchmark Models

The final controlled lookback grid is $L \in \{4, 8, 12\}$. Let $\mathcal{M} = \{\text{Technical}, \text{Technical+HMM}\}$ denote the set of feature configurations, and let $\mathcal{S} = \{7, 19, 42, 91, 123\}$ denote the set of random seeds. For every ticker, feature configuration, lookback, and random seed, a separate LSTM is trained. For ticker i and feature configuration $m \in \mathcal{M}$, the selected lookback is

$$L^*_{i,m} = \arg \max_{L\in\{4,8,12\}} \frac{1}{|\mathcal{S}|} \sum_{s\in\mathcal{S}} \text{AUC}^{\text{validation}}_{i,m,L,s}.$$

Here, $\text{AUC}^{\text{validation}}_{i,m,L,s}$ denotes the validation ROC-AUC obtained for ticker i, feature configuration m, lookback L, and random seed s. Shorter lookbacks are preferred when the mean validation ROC-AUC values are tied. Because the HMM feature may alter the validation surface, the two feature configurations are allowed to select different lookbacks.

At the selected lookback, test metrics are reported as the mean and standard deviation across the five seed-level LSTM fits. The controlled results are therefore not based on a single favourable random seed. For the displayed ROC and precision-recall curves, the five seed-level probability forecasts are averaged to produce one curve for each ticker and model.

After the probability model is fixed, a classification threshold is selected separately for each ticker, feature configuration, and seed-level LSTM fit. Let $\mathcal{G}_\tau = \{0.35, 0.36, \ldots, 0.65\}$ denote the candidate threshold grid. For ticker i, feature configuration $m \in \mathcal{M}$, and random seed $s \in \mathcal{S}$, the selected threshold is

$$\tau^*_{i,m,s} = \arg \max_{\tau\in\mathcal{G}_\tau} \text{BA}^{\text{validation}}_{i,m,s}(\tau).$$

Here, BA denotes balanced accuracy, whose formal definition is provided in Section 4.5. Ties are resolved by choosing the threshold closest to 0.50 and, if a tie remains, the smaller threshold. Test outcomes are not used to select the lookback, classification threshold, architecture, or other training settings.

The controlled comparison includes six model families:

1. **Training-majority baseline.** This model assigns the positive-class share observed in the training sample as a constant probability to every observation and applies a threshold of 0.50. Because the positive-class share in the training sample exceeds 0.50 for all three tickers, the model predicts the up class for every observation.

2. **Persistence baseline.** This model predicts that the next week's direction is equal to the current week's direction. Its score is $\mathbf{1}\{r_{i,w} > 0\}$.

3. **HMM-only probability benchmark.** This model uses `HMM_Up_Prob_Raw` directly as the next-week probability score. Its classification threshold is selected to maximize validation balanced accuracy.

4. **Technical logistic regression.** This model uses the 15 standardized technical variables from the current week rather than an LSTM sequence. It is estimated separately for each ticker without class weighting. The inverse regularization strength is selected from $C \in \{0.01, 0.1, 1.0, 10.0\}$ using validation ROC-AUC, with the smaller value preferred in the event of a tie. Smaller values of C correspond to stronger regularization. The maximum number of iterations is 5000. After selecting C, the logistic-regression classification threshold is selected from the same 0.35–0.65 grid by maximizing validation balanced accuracy, using the same tie-breaking rule described above.

5. **Technical-only LSTM.** This is the nonlinear sequence benchmark using only the 15 technical predictors.

6. **Technical-plus-HMM LSTM.** This is the primary hybrid model. It uses the same architecture and technical predictors as the technical-only LSTM, with the addition of `HMM_Up_Prob_Scaled`.

These baselines separate several questions. The majority baseline measures the effect of class imbalance, persistence tests simple directional continuation, the HMM-only model measures the predictive content of the HMM score without an LSTM, logistic regression tests whether a simpler linear classifier is sufficient, and the paired LSTMs isolate the incremental contribution of the HMM-derived feature.

4.5 Chronological Evaluation, Leakage Controls, and Metrics

Several controls are implemented to prevent future information from entering model estimation. First, all splits are chronological, ticker-specific, and assigned according to the target date. Second, technical-feature scalers are fitted separately by ticker using training observations only. Third, each fixed-split HMM and its observation scaler are estimated using training observations only. Fourth, the HMM state labels and state-specific positive-return frequencies are calculated from training observations only. Fifth, the HMM feature supplied to the LSTM is standardized using a separate scaler fitted to training observations only. Finally, validation observations are used only for model and threshold selection, while test observations are reserved for final evaluation.

A validation or test sequence may include previous input observations that belong to a preceding split. This does not introduce look-ahead bias because those observations would already have been available at the target date. Sequence membership is determined by the target row, and the target date is always later than every date in the corresponding input sequence.

ROC-AUC is used to measure threshold-independent ranking performance. Precision-recall AUC is reported as a complementary, prevalence-sensitive metric focusing on the model's ability to identify positive-return weeks.

Balanced accuracy is the principal threshold-dependent metric because it gives equal weight to the positive and negative classes:

$$\text{BA} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right).$$

Here, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. The positive class corresponds to an up week, while the negative class corresponds to a non-positive week.

Raw accuracy, precision, recall, F1 score, Cohen's kappa, confusion-matrix counts, predicted up share, and actual up share are also recorded to diagnose class-specific behaviour. The F1 score is not interpreted alone as evidence of discrimination because a model can achieve a high F1 score by frequently predicting the majority positive class.

To quantify uncertainty in the incremental contribution of the HMM feature, two-sided 95% percentile paired circular moving-block bootstrap confidence intervals are calculated from the saved seed-level test predictions. For each block length, 10,000 bootstrap replications are generated by sampling overlapping consecutive test-week blocks with replacement, wrapping blocks around the end of the test sequence when necessary. The same sampled indices are applied to the technical-only and technical-plus-HMM predictions. Within each replication, the paired metric difference is calculated for each matched random seed and then averaged across the five seeds. The interval endpoints are the empirical 2.5th and 97.5th percentiles of this bootstrap distribution.

Block lengths of 4, 8, and 12 weeks are used as a sensitivity analysis. The classification thresholds remain fixed at the values selected from the validation sample. The resulting intervals therefore describe test-period uncertainty conditional on the fitted models and selected hyperparameters; they do not include the uncertainty associated with retraining or repeating the complete model-selection procedure.

4.6 Design of Secondary and Robustness Analyses

Several secondary analyses examine whether the primary conclusion is sensitive to broader market information, alternative HMM specifications, target alignment, or the choice of evaluation period. Volatility-context experiments add VIX and VXN variables to test whether market-wide and technology-sector uncertainty provides information beyond stock-specific technical features. Compact and fuller volatility specifications are evaluated both with and without HMM features. The market-context experiments add QQQ variables and the stock's return relative to QQQ. Larger exploratory specifications also include recession/expansion state information. These experiments test whether stock-specific HMM information remains useful once broader market conditions are represented directly.

The HMM robustness analysis compares the primary univariate two-state HMM with richer two-state and three-state specifications. These variants assess whether the conclusions depend strongly on the number of states or the set of HMM observations.

Target-overlap and calibration diagnostics compare the distributions of the HMM-implied up probabilities across observed target classes and across probability bins. Their purpose is to distinguish whether the HMM provides an economically interpretable regime description from whether it is well aligned with the next-week classification target. Because repeating the complete expanding-window procedure for all three tickers was beyond the available project scope, the rolling replication was restricted to AAPL and IBM. These tickers provide contrasting fixed-split cases: adding the HMM feature produced a clear deterioration for AAPL, whereas IBM showed a small positive ROC-AUC difference but lower balanced accuracy. The rolling analysis is therefore interpreted as a targeted robustness exercise rather than a complete three-ticker replication. Consequently, no conclusion about the temporal stability of the HMM effect for MSFT can be drawn from this analysis.

Finally, a rolling paired replication is conducted for AAPL and IBM using nine expanding-window folds. Each fold contains a 52-week validation period and a 52-week test period, with a 52-week step between folds. HMMs, technical scalers, HMM-feature scalers, model selection, and threshold selection are repeated within each fold using only information available at that stage.

Rolling replication is interpreted as secondary robustness evidence rather than an exact repetition of the primary three-ticker design. AAPL uses technical and volatility-context base variables, whereas IBM uses technical base variables. In both cases, the paired HMM variant adds the fold-specific scaled HMM-implied probability.

The numerical findings, tables, and figures of these secondary experiments are reported separately from the primary controlled results. The best isolated result from a broad exploratory grid is not treated as evidence that the corresponding model is generally superior.

5 HMM Regime Diagnostics

Before evaluating forecast performance, the fitted HMM regimes are descriptively examined to assess whether the inferred states correspond to economically plausible market conditions. The weekly HMM is fitted separately for each ticker as a two-state Gaussian model using only `Weekly_Log_Return` as the observed variable. The scaler and HMM are fitted on training rows only. The plotted probabilities are causal filtered probabilities $\Pr(S_{i,w} \mid \widetilde{r}_{i,1:w})$, rather than smoothed probabilities that use future observations. States are labelled using training-period mean weekly returns: the higher-mean state is labelled bullish and the lower-mean state bearish.

Table 4: Training-period weekly HMM regime profiles.

Ticker	Regime	Count	Mean return	Volatility	Positive share	Expected duration
AAPL	Bullish	332	0.010	0.022	0.678	10.6
AAPL	Bearish	233	-0.003	0.053	0.455	9.5
IBM	Bullish	430	0.002	0.017	0.535	11.0
IBM	Bearish	135	-0.003	0.055	0.489	4.6
MSFT	Bullish	490	0.006	0.021	0.604	12.2
MSFT	Bearish	75	-0.004	0.065	0.453	3.6

Table 4 shows that the labelled bullish states have higher training-period mean returns and higher positive-return frequencies. However, the volatility differences are larger than the mean-return differences for all three tickers. The regimes should therefore be interpreted cautiously as relative bull/bear or calm/stress states rather than exact historical bull and bear market classifications.

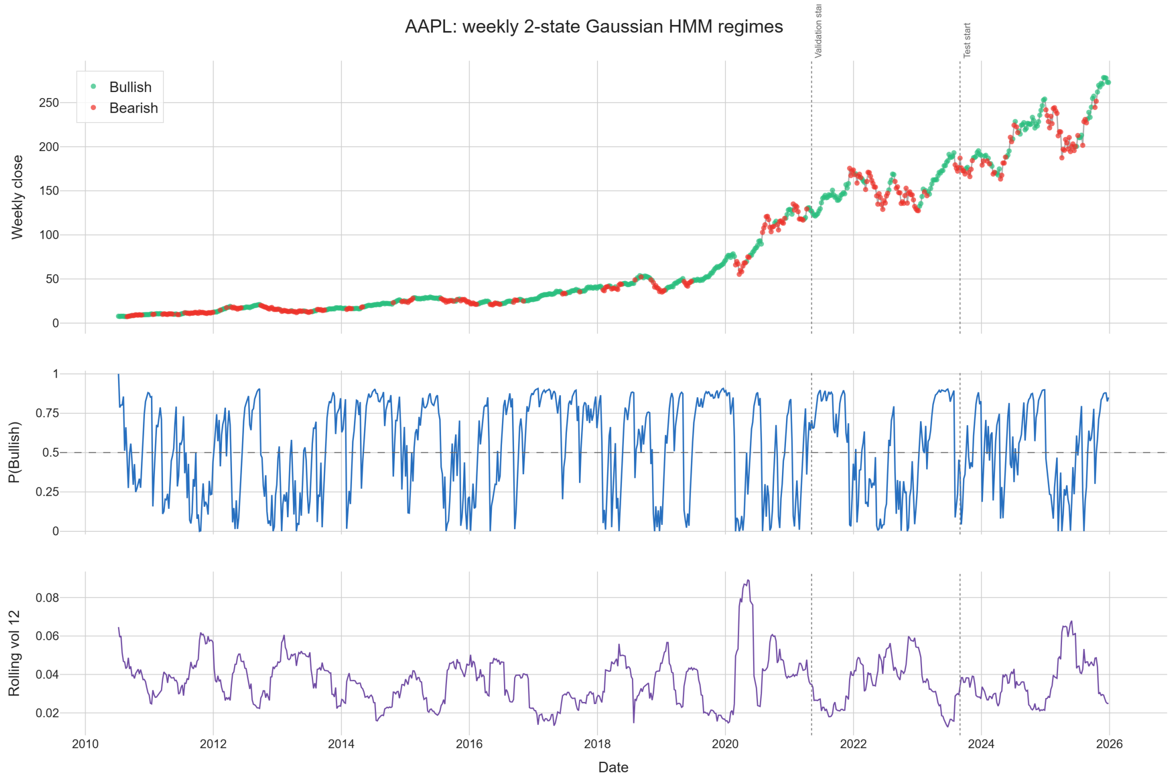


Figure 3: Weekly two-state Gaussian HMM regime diagnostics for AAPL. Green and red observations indicate the training-labelled bullish and bearish regimes. The middle panel shows the causal filtered bullish-state probability, while the lower panel shows 12-week rolling volatility. Dashed vertical lines indicate the validation and test starts.

Bearish or stress-state probabilities rise plausibly around recognised stress episodes, including the early-2020 COVID shock. This is an economic plausibility check, not evidence of forecasting usefulness. Predictive value is evaluated separately through the HMM-only benchmark and the controlled technical-only versus technical-plus-HMM comparison.

6 Primary Results: Controlled HMM Feature Comparison

The primary empirical analysis evaluates whether adding the scaled HMM-implied next-week up probability, `HMM_Up_Prob_Scaled`, improves an LSTM that already receives the 15 stock-specific technical predictors. The technical-only and technical-plus-HMM LSTMs use the same target observations, chronological splits, candidate lookback grid, architecture, training procedure, random seeds, and validation-based selection rules, so the only systematic input difference is the additional HMM-derived probability. The selected lookbacks were four weeks for the AAPL technical-only LSTM and 12 weeks for its HMM-informed counterpart, while both IBM models and both MSFT models selected four weeks; technical logistic regression selected $C = 0.01$ for AAPL and IBM and $C = 0.10$ for MSFT. Although 0.50 is the conventional sigmoid threshold, every seed-level LSTM was classified using its own validation-selected threshold, and the reported threshold values are therefore summaries rather than universal decision boundaries: the mean technical-only and technical-plus-HMM thresholds were 0.568 and 0.602 for AAPL, 0.522 and 0.544 for IBM, and 0.534 and 0.554 for MSFT.

Table 5: Final controlled test performance. LSTM values are reported as mean ± standard deviation across five seed-level fits; deterministic benchmark values are single test-sample estimates.

Ticker	Model	ROC-AUC	PR-AUC	Balanced accuracy	F1 score
AAPL	HMM-only probability	0.437	0.513	0.451	0.355
AAPL	Persistence	0.469	0.535	0.469	0.480
AAPL	Technical logistic regression	**0.583**	**0.624**	**0.547**	0.486
AAPL	Technical-only LSTM	0.531 ± 0.040	0.577 ± 0.046	0.525 ± 0.029	0.531 ± 0.128
AAPL	Technical-plus-HMM LSTM	0.470 ± 0.011	0.543 ± 0.031	0.480 ± 0.030	0.326 ± 0.075
AAPL	Training-majority	0.500	0.549	0.500	0.709
IBM	HMM-only probability	0.539	0.616	0.553	0.763
IBM	Persistence	0.516	0.614	0.516	0.607
IBM	Technical logistic regression	0.580	0.673	**0.555**	0.380
IBM	Technical-only LSTM	0.583 ± 0.046	0.680 ± 0.022	0.552 ± 0.050	0.430 ± 0.139
IBM	Technical-plus-HMM LSTM	$\mathbf{0.585} \pm 0.028$	$\mathbf{0.685} \pm 0.014$	0.532 ± 0.017	0.313 ± 0.073
IBM	Training-majority	0.500	0.607	0.500	0.755
MSFT	HMM-only probability	0.453	0.567	0.488	0.706
MSFT	Persistence	**0.542**	0.588	**0.542**	0.569
MSFT	Technical logistic regression	0.415	0.561	0.413	0.564
MSFT	Technical-only LSTM	0.476 ± 0.020	$\mathbf{0.593} \pm 0.033$	0.472 ± 0.014	0.592 ± 0.137
MSFT	Technical-plus-HMM LSTM	0.474 ± 0.040	0.566 ± 0.024	0.466 ± 0.044	0.528 ± 0.131
MSFT	Training-majority	0.500	0.566	0.500	0.723

Note: Bold values identify the strongest mean or single test-sample estimate for ROC-AUC, PR-AUC, and balanced accuracy for each ticker. F1 score is reported as a complementary classification metric but is not used to identify the preferred model, because it is strongly influenced by the majority positive class.

Table 5 shows that greater model complexity was not consistently rewarded, because technical logistic regression achieved the highest balanced accuracy for AAPL and IBM, whereas the persistence baseline performed best for MSFT. For AAPL, technical logistic regression achieved ROC-AUC 0.583 and balanced accuracy 0.547, while adding the HMM probability reduced the LSTM ROC-AUC from 0.531 to 0.470 and balanced accuracy from 0.525 to 0.480; for IBM, the HMM-informed LSTM produced the highest ROC-AUC and PR-AUC, but its gains over the technical-only LSTM were only 0.002 and 0.005, and its balanced accuracy was lower; and for MSFT, both learned LSTM configurations remained below 0.50 in ROC-AUC, while persistence reached 0.542, which indicates that the engineered technical and HMM features did not provide stable ranking information during the controlled test period.

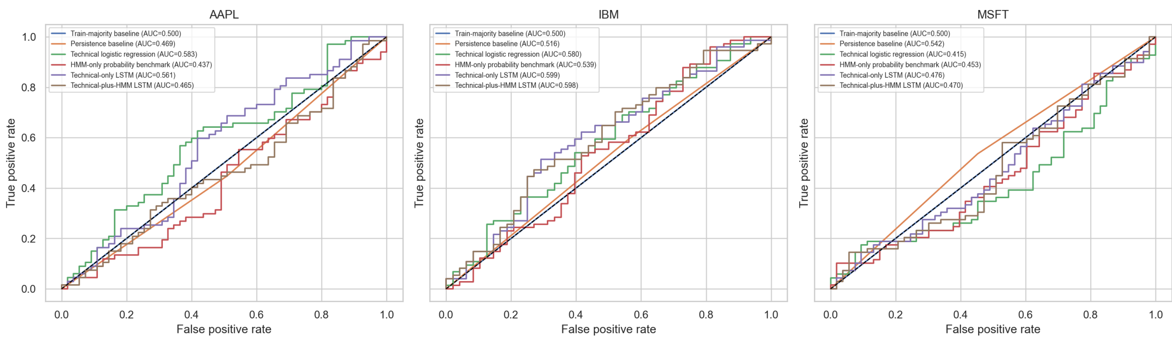


Figure 4: Controlled test ROC curves for the benchmark, logistic-regression, HMM-only, and LSTM-based models. For the LSTM models, the curves are constructed from probabilities averaged across the five selected seed-level fits, whereas Table 5 reports the mean of the five seed-level ROC-AUC values; consequently, the AUC displayed in the figure may differ slightly from the tabulated mean ROC-AUC.

The direct incremental comparison is defined as $\Delta M_i = M_{i,\text{Technical+HMM}} - M_{i,\text{Technical}}$, so a positive value indicates that the HMM feature improves the corresponding metric and a negative value indicates deterioration. For AAPL, ΔROC-AUC $= -0.061$, ΔPR-AUC $= -0.035$, and ΔBA $= -0.045$; for IBM, the respective differences are $+0.002$, $+0.005$, and -0.020; and for MSFT they are -0.003, -0.027, and -0.006. Because very small differences can arise from the particular test period and trained models, paired moving-block bootstrap intervals were calculated from the saved seed-level test predictions, while the same resampled test-week indices were applied to each matched technical-only and technical-plus-HMM model.

Table 6: Two-sided 95% percentile paired circular moving-block bootstrap intervals for the incremental HMM effect, using saved seed-level test predictions.

Ticker	Metric	Delta	4-week CI	8-week CI	12-week CI
AAPL	ROC-AUC	-0.061	[-0.112,-0.011]	[-0.111,-0.010]	[-0.107,-0.014]
AAPL	Balanced accuracy	-0.045	[-0.080,-0.009]	[-0.079,-0.011]	[-0.075,-0.013]
IBM	ROC-AUC	0.002	[-0.029,0.035]	[-0.027,0.034]	[-0.026,0.033]
IBM	Balanced accuracy	-0.020	[-0.050,0.008]	[-0.047,0.006]	[-0.044,0.003]
MSFT	ROC-AUC	-0.003	[-0.031,0.028]	[-0.029,0.030]	[-0.025,0.026]
MSFT	Balanced accuracy	-0.006	[-0.040,0.029]	[-0.042,0.031]	[-0.042,0.030]

The bootstrap results reinforce the descriptive comparison, because the AAPL intervals remain below zero for both ROC-AUC and balanced accuracy under all three block-length assumptions, whereas every IBM and MSFT interval contains zero. The controlled evidence therefore supports a clear deterioration for AAPL, but it does not establish either an improvement or a deterioration for IBM and MSFT; in particular, IBM's positive ROC-AUC difference of 0.002 is too small and uncertain to be treated as a meaningful ranking advantage, while MSFT remains essentially unchanged to slightly worse. Overall, the primary controlled comparison does not support the claim that the isolated HMM-implied probability reliably improves an otherwise comparable technical LSTM.

6.1 Exploratory Best-Observed Market-Context Models

Although the primary controlled comparison did not find consistent incremental value from adding the isolated HMM-derived probability, the broader exploratory grid produced stronger best-observed ranking performance when market-wide and volatility-related variables were included. Figure 5 reports the highest test ROC-AUC identified for each ticker across the wider set of feature configurations, lookback lengths, and stacked LSTM architectures, with AAPL achieving 0.609 using volatility, market, QQQ, and HMM features with a 12-week lookback, while IBM and MSFT achieved 0.699 and 0.663, respectively, using volatility, market, and QQQ features with eight-week lookbacks.

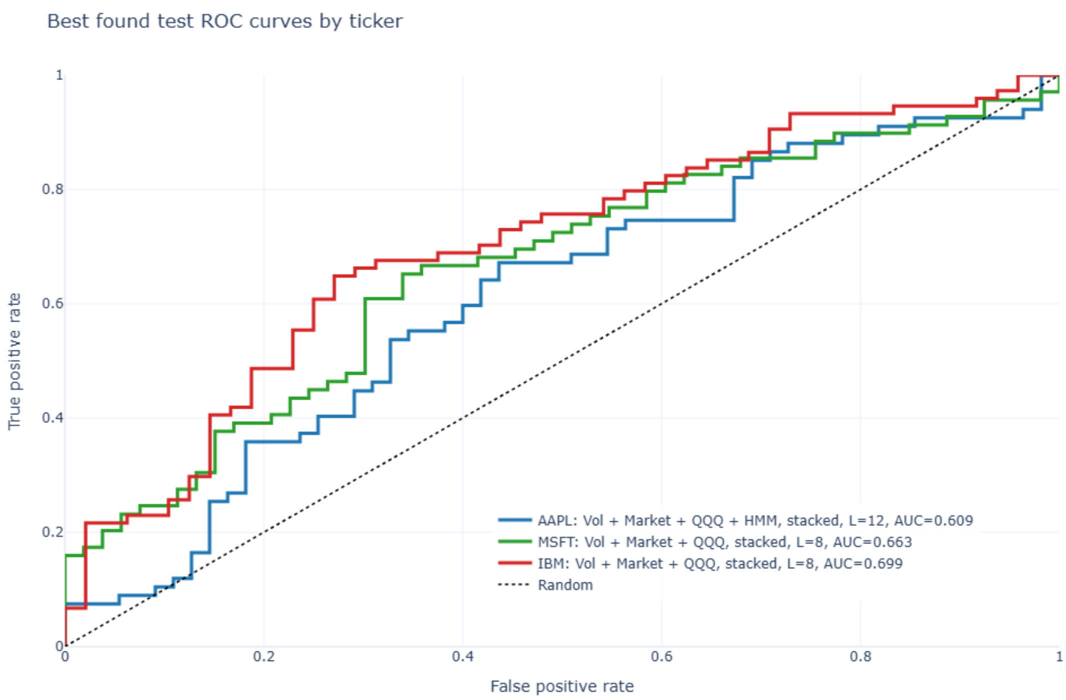


Figure 5: Best-observed test ROC curves by ticker across the broader exploratory model grid. AAPL's strongest configuration included volatility, market, QQQ, and stock-HMM features with a stacked 12-week LSTM, whereas the strongest IBM and MSFT configurations used volatility, market, and QQQ inputs with stacked eight-week LSTMs. Because the configurations were identified after inspecting multiple test-result combinations, the reported ROC-AUC values are exposed to test-set selection and multiple-comparison optimism.

The strongest exploratory AAPL specification included the stock-specific HMM feature, whereas the strongest IBM and MSFT specifications did not, so the broader results do not indicate a universal benefit from regime information. The MSFT contrast is particularly notable because the controlled technical-only and technical-plus-HMM LSTMs obtained ROC-AUC values of 0.476 and 0.474, while the broader volatility-plus-market-plus-QQQ configuration achieved 0.663; this pattern is consistent with the possibility that common market conditions contain information not captured by the compact stock-specific feature set, although it does not establish that the exploratory configuration would generalise to new data. These findings therefore do not overturn the controlled conclusion, because the best-observed configurations were selected retrospectively using their test results, but they identify market-context feature families that warrant future evaluation through a validation-selected or nested rolling procedure in which the final test periods remain untouched during model selection.

7 Rolling Paired Replication Results

To examine whether the fixed-split findings remain stable across different market periods, a rolling paired replication was completed for AAPL and IBM using nine annual expanding-window folds. Each fold contains a 52-week validation period and a 52-week test period, while the starting point advances by 52 weeks between consecutive folds. Within every fold, the HMM and all feature scalers are re-estimated using fold-training observations only, lookback lengths and classification thresholds are selected using fold-validation data, and five matched LSTM seeds are averaged to produce ensemble probability forecasts for the fold-test period. The paired comparison is performed on identical observations within each fold, although the base feature set differs by ticker: the AAPL base model contains technical and volatility-context variables, whereas the IBM base model contains the technical variables only; in both cases, the HMM-informed specification adds the fold-specific scaled HMM-implied next-week up probability.

Table 7: Rolling paired HMM replication summary.

Ticker	Folds	Mean base AUC	Mean HMM AUC	Mean delta	Pooled base AUC	Pooled HMM AUC
AAPL	9	0.479	0.490	0.010	0.437	0.434
IBM	9	0.545	0.527	-0.018	0.512	0.509

Table 7 shows that AAPL has a small positive mean fold-level ROC-AUC difference of 0.010, although the HMM-informed model improves upon the base model in only four of the nine folds, whereas IBM has a negative mean difference of -0.018 and improves in only two folds. The instability of the incremental HMM effect is illustrated in Figure 6, where both tickers display positive and negative differences across the evaluation periods; although several individual folds show noticeable improvements, particularly Fold 7 for AAPL and Fold 4 for IBM, these gains are offset by deterioration in other periods and therefore do not provide evidence of a persistent forecasting advantage.

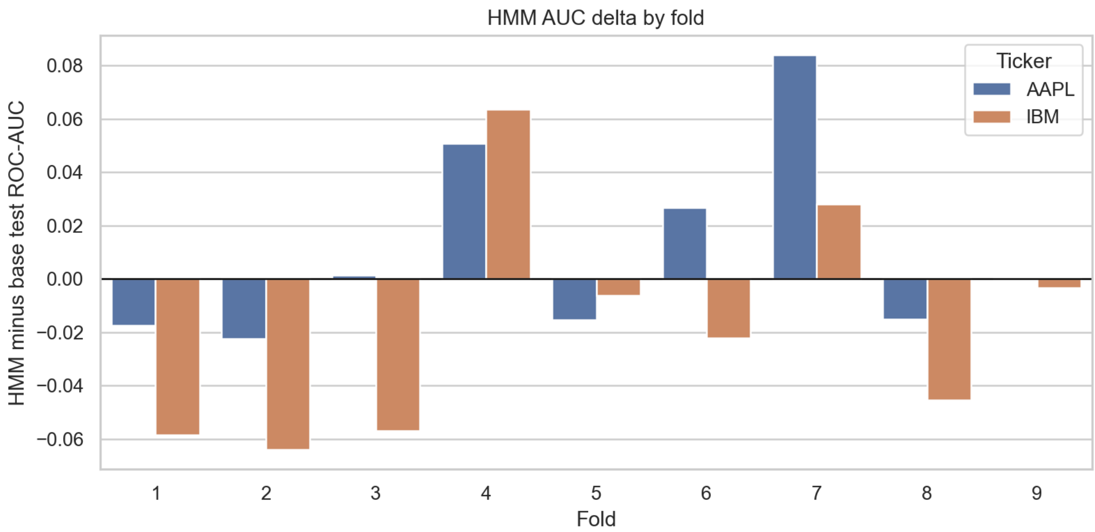
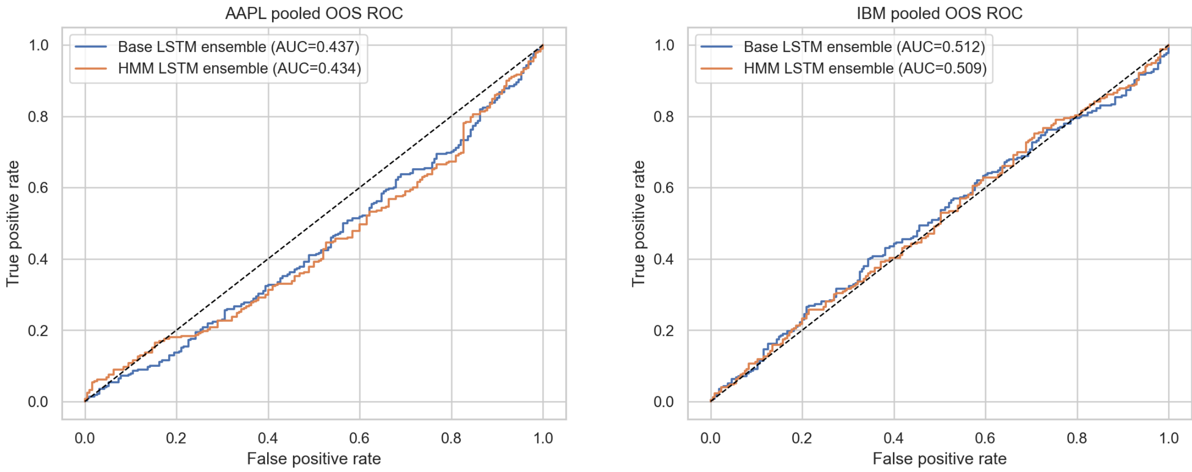


Figure 6: Fold-level change in test ROC-AUC after adding the fold-specific HMM-implied up probability to the corresponding base LSTM. Positive values indicate an improvement from the HMM feature, whereas negative values indicate deterioration.

The pooled out-of-sample results provide a complementary assessment because they combine all test observations across the nine folds and evaluate the overall ranking of positive and non-positive weeks. Figure 7 shows that the pooled ROC-AUC decreases slightly from 0.437 to 0.434 for AAPL and from 0.512 to 0.509 for IBM after the HMM feature is added. AAPL's positive mean fold-level difference and negative pooled difference are not mathematically contradictory, because the mean fold AUC assigns equal weight to each fold's within-period ranking performance, whereas the pooled AUC ranks all out-of-sample observations jointly and may also be affected by differences in score distributions across models that were independently refitted in each fold.



(a) AAPL **(b) IBM**

Figure 7: Pooled rolling out-of-sample ROC curves for the base and HMM-informed LSTM ensembles. The predictions from all nine annual test folds are combined, while the HMM, feature scalers, lookback selection, threshold selection, and LSTM estimation are repeated separately within every fold.

Exact two-sided sign-flip permutation tests were applied to the nine paired fold-level ROC-AUC differences, producing $p = 0.477$ for AAPL and $p = 0.242$ for IBM. These results do not provide evidence of a reliable rolling HMM effect, although the inference is necessarily limited by the small number of folds and should not be interpreted as proof that the incremental effect is exactly zero. Taken together, the fold-level variation, pooled out-of-sample curves, and permutation results strengthen the primary fixed-split conclusion: the HMM-derived probability can improve ranking during particular market periods, but its contribution is neither sufficiently frequent nor sufficiently stable to establish consistent predictive value beyond the corresponding base LSTM.

8 Discussion

The HMM produced an economically interpretable description of latent market conditions, because the training-labelled bullish states had higher mean weekly returns and positive-return frequencies, whereas the bearish states were associated with substantially higher volatility and, particularly for IBM and MSFT, shorter expected durations. These differences support interpreting the estimated states as relative bullish/bearish or calm/stress regimes, although they should not be treated as exact classifications of historically recognised bull and bear markets. The regime diagnostics therefore establish descriptive usefulness, but they do not by themselves establish predictive usefulness, because a state that summarises the current return and volatility environment may still have little stable relationship with the sign of the following weekly return.

The controlled feature-ablation results show that this distinction is empirically important. For AAPL, adding the HMM-implied probability reduced LSTM ROC-AUC by

0.061 and balanced accuracy by 0.045, while the paired moving-block bootstrap intervals remained below zero for all examined block lengths; consequently, the evidence supports a genuine deterioration conditional on the fitted models and selected hyperparameters. For IBM, the HMM-informed LSTM increased ROC-AUC by only 0.002 and PR-AUC by 0.005, while balanced accuracy declined by 0.020, and the corresponding bootstrap intervals included zero, so the apparent ranking gain should be treated as practically indistinguishable from no effect. MSFT showed similarly weak evidence, because both LSTM variants had ROC-AUC below 0.50, the incremental differences were close to zero, and the bootstrap intervals again included zero. The stronger performance of technical logistic regression for AAPL and IBM, together with the persistence baseline for MSFT, further indicates that the additional flexibility of the LSTM and HMM–LSTM models was not consistently rewarded in this weekly small-sample setting.

The completed rolling replication provides a second explanation for why the early HMM–LSTM findings appeared more favourable than the final conclusion. Initial experiments searched for several lookbacks, feature sets, architectures, and thresholds, and some same-lookback comparisons produced encouraging balanced-accuracy improvements; however, the final controlled analysis used matched random seeds, validation-only selection, leakage-aware HMM features, and stronger benchmarks, while the rolling analysis repeated estimation across nine annual test periods. AAPL obtained a small positive mean fold-level ROC-AUC difference of 0.010, but the HMM variant improved in only four of nine folds and its pooled ROC-AUC was slightly lower, whereas IBM obtained a negative mean difference of -0.018, improved in only two folds, and also had a lower pooled ROC-AUC. The exact sign-flip permutation results, with $p = 0.477$ for AAPL and $p = 0.242$ for IBM, do not support a stable paired HMM effect, although the limited number of folds restricts inferential power. The difference between AAPL's positive mean fold effect and negative pooled effect is not contradictory, because mean fold AUC averages within-period rankings equally, whereas pooled AUC ranks all out-of-sample observations jointly and is influenced by differences in probability scales across models that are refitted separately in each fold.

The broader exploratory results nevertheless suggest that market context may be more informative than the isolated HMM probability. Best-observed ROC-AUC values reached 0.609 for AAPL, 0.699 for IBM, and 0.663 for MSFT when volatility and market-context variables were included, while the strongest IBM and MSFT specifications did not contain the stock-specific HMM feature. The AAPL best-observed configuration included both HMM and market-context variables, which raises the possibility that regime information may interact with a richer representation of market conditions; however, these configurations were identified retrospectively from a broad test-result grid, so they are exposed to multiple-comparison and test-set-selection optimism and cannot overturn the controlled and rolling conclusions.

A useful methodological reference is the high-frequency LSTM ensemble study of Borovkova and Tsiamas [7], although its results are not directly comparable with those of the present project. Their analysis uses one year of five-minute observations from

2014, producing approximately 18,252 observations per stock, and evaluates dynamically weighted ensembles of 12 stacked LSTMs across 22 large-cap US stocks and 21 rolling prediction periods. The stock universes also differ, with IBM providing one point of overlap, while AAPL and MSFT are not among their predicted stocks. By contrast, the present analysis uses weekly observations over approximately 2010–2025 and therefore covers a much longer and more heterogeneous set of market conditions, although it contains far fewer observations for neural-network estimation. The two studies should consequently be viewed as complementary: Borovkova and Tsiamas examine whether large adaptive ensembles can identify persistent five-minute patterns from a high-frequency sample, whereas this project examines whether a causal HMM-derived regime probability adds stable next-week predictive information across a substantially longer historical period.

Taken together, the results indicate that the HMM's most reliable contribution is descriptive rather than predictive. The HMM identifies persistent and economically interpretable changes in return and volatility conditions, but the resulting next-week up probability does not provide stable incremental information once the technical LSTM, validation procedure, test observations, and random seeds are held constant. The project therefore contributes a negative but informative result: an interpretable latent regime can be useful for understanding market conditions without necessarily improving next-week directional classification.

9 Limitations

The first limitation is the relatively small effective sample size. Weekly aggregation changes the prediction horizon and reduces the number of observations available for estimation, so each ticker provides only several hundred training sequences after rolling-feature warm-up and lookback construction. This setting increases the risk that a flexible LSTM learns unstable relationships, while the five random seeds measure optimisation variation rather than the full sampling uncertainty of the model-selection procedure. The moving-block bootstrap improves the uncertainty assessment by preserving short-range dependence in the test weeks, but its intervals remain conditional on the fitted HMMs, selected lookbacks, validation-selected thresholds, and stored neural models, because the complete training and model-selection pipeline is not repeated within every bootstrap sample.

The second limitation is external validity, because AAPL, IBM, and MSFT are large, liquid, technology-related companies and therefore do not represent the wider equity market. Although rolling replication evaluates nine annual periods, it is restricted to AAPL and IBM, and the base specifications are not identical across the two tickers because AAPL uses technical and volatility-context variables while IBM uses the technical feature set. The rolling folds also share expanding training histories, so their paired differences should not be interpreted as fully independent observations. These design choices provide useful robustness evidence, but they do not establish that the conclusions generalise to other securities, sectors, frequencies, or market regimes.

The third limitation concerns model and feature selection. Several HMM specifications, feature groups, lookbacks, thresholds, and LSTM architectures were examined during the exploratory stage, and the best-observed context models were identified after inspecting their test ROC-AUC values. Their reported maxima are therefore likely to be optimistic, while a genuinely confirmatory evaluation would require nested validation or an untouched sequence of future test periods. Finally, the analysis evaluates statistical classification performance rather than economic profitability, because ROC-AUC, PR-AUC, balanced accuracy, and F1 score do not incorporate transaction costs, bid–ask spreads, slippage, turnover, position sizing, or risk-adjusted returns; therefore, even a model with above-random directional ranking would not automatically constitute a profitable trading strategy. A further limitation concerns neural-network interpretability. An Integrated Gradients analysis was planned to examine how technical, market-context, and HMM-derived variables contributed across features and weekly lags, but this analysis was not completed within the project scope. Consequently, the report evaluates whether the additional HMM feature changes predictive performance, but it does not determine how the fitted LSTMs internally use that feature or whether its contribution varies across observations and time steps.

10 Conclusion

This honours project examined whether an HMM-derived regime probability improves LSTM-based prediction of next-week stock direction beyond stock-specific technical features for AAPL, IBM, and MSFT. The project began with daily next-day return regression, but the early models produced low-variance predictions and weak directional discrimination, so the final task was reformulated as weekly binary classification. A two-state Gaussian HMM was then used to construct causal regime probabilities, while the primary predictive analysis compared technical-only and technical-plus-HMM LSTMs under matched data splits, architectures, random seeds, validation rules, and benchmark models.

The HMM diagnostics showed that the estimated states were economically interpretable, because lower-return regimes were also associated with higher volatility and different persistence patterns; however, the controlled predictive evidence did not show a reliable incremental HMM benefit. Adding the HMM probability clearly reduced AAPL performance, while IBM exhibited only a negligible ranking improvement accompanied by weaker balanced accuracy, and MSFT remained close to or below random ranking. The paired moving-block bootstrap confirmed that the AAPL deterioration was robust across the examined block lengths, whereas the IBM and MSFT differences remained compatible with no incremental effect.

The completed rolling replication strengthened this interpretation across multiple market periods. AAPL obtained a small positive average fold-level difference but improved in fewer than half of the folds and had a slightly lower pooled ROC-AUC, whereas IBM showed a negative average difference, improved in only two folds, and also had a lower pooled ROC-AUC. Exact sign-flip tests did not provide evidence of a stable paired effect for either ticker. Broader exploratory market-context models produced higher best-observed ROC-AUC values, but those configurations were retrospectively selected from the test results and must therefore remain hypothesis-generating rather than confirmatory.

The final conclusion is consequently cautious but clear: HMMs provide a useful and interpretable description of latent weekly market conditions, while LSTMs offer a flexible framework for sequential classification, but combining the two did not consistently improve next-week directional forecasts in the completed controlled and rolling experiments. Future research should evaluate market-context and regime features through a nested rolling design across a larger and more diverse stock universe, while preserving untouched test periods and adding an explicit trading evaluation that accounts for costs and risk.

A Additional Weekly HMM Diagnostics

The main text presents AAPL as the primary visual example of the two-state weekly HMM. The corresponding IBM and MSFT results are reported below to show how the inferred regimes differ across the three stocks. In every ticker, the state with the higher training-period mean weekly return is labelled bullish, while the lower-return state is labelled bearish.

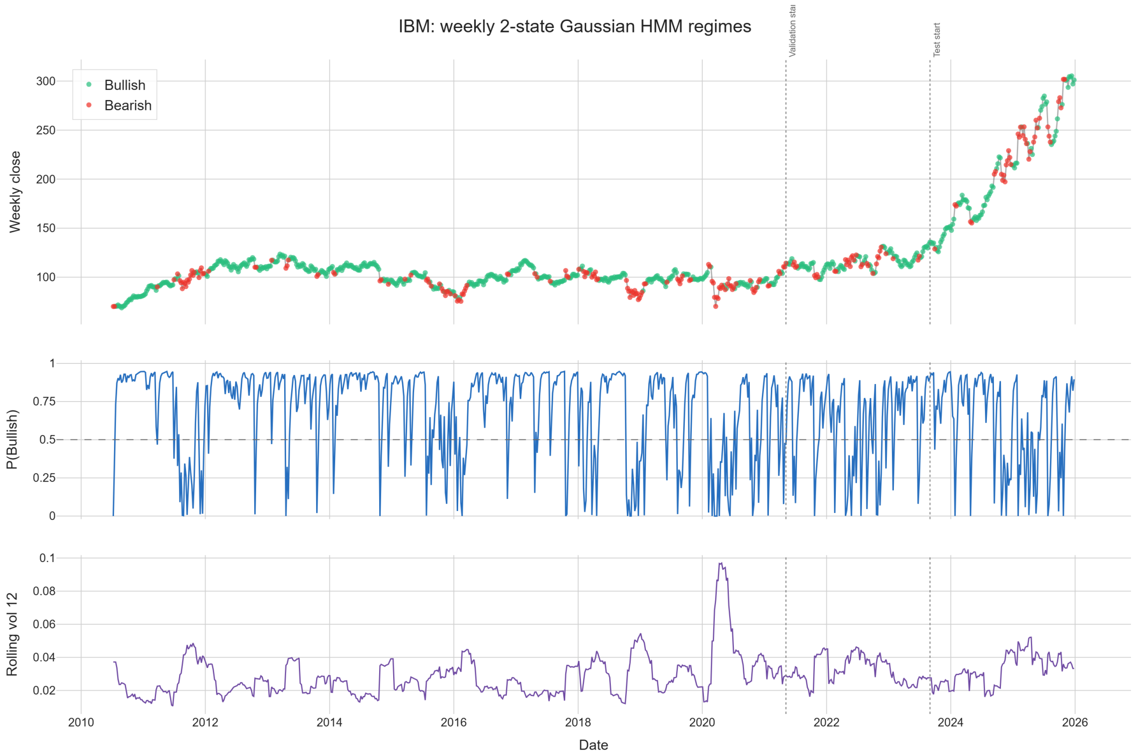


Figure 8: IBM weekly closing price, filtered bullish-state probability, and 12-week rolling volatility under the two-state Gaussian HMM. Vertical lines indicate the beginning of the validation and test periods.

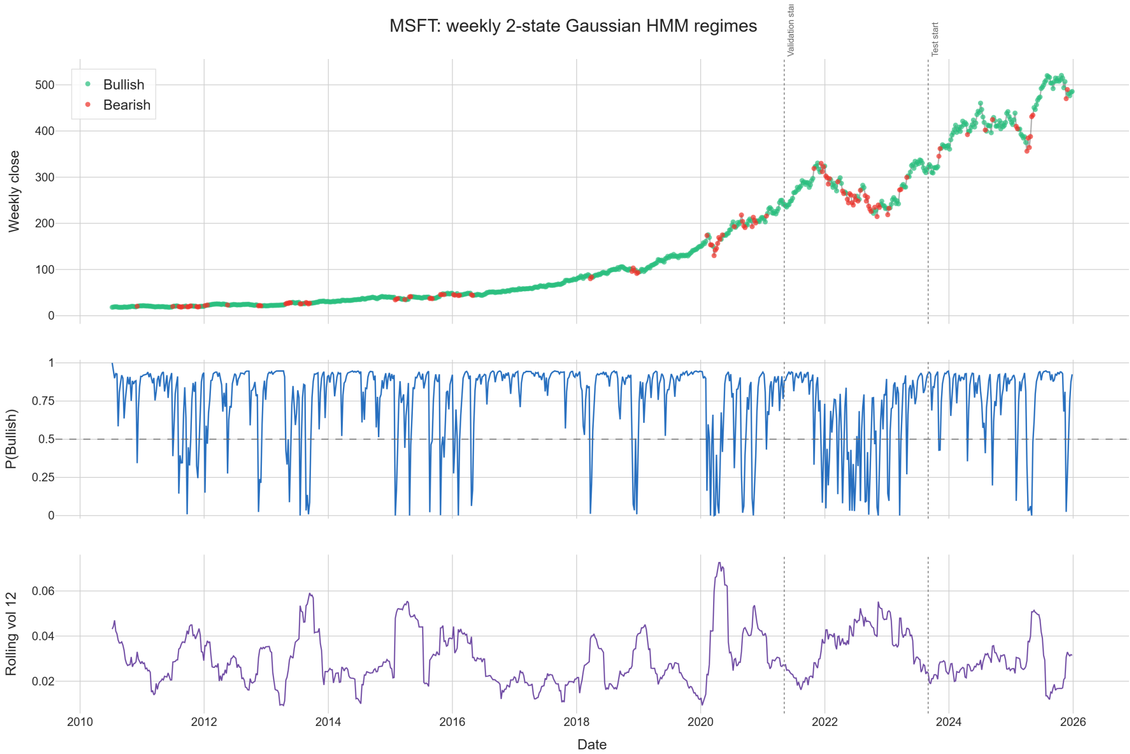


Figure 9: MSFT weekly closing price, filtered bullish-state probability, and 12-week rolling volatility under the two-state Gaussian HMM. Vertical lines indicate the beginning of the validation and test periods.

Figure 10 compares the distribution of the HMM-derived probabilities conditional on the realised next-week direction. The substantial overlap between bearish and bullish observations shows that neither the regime probability nor the derived positive-return probability cleanly separates the two target classes, even when the HMM states themselves remain economically interpretable.

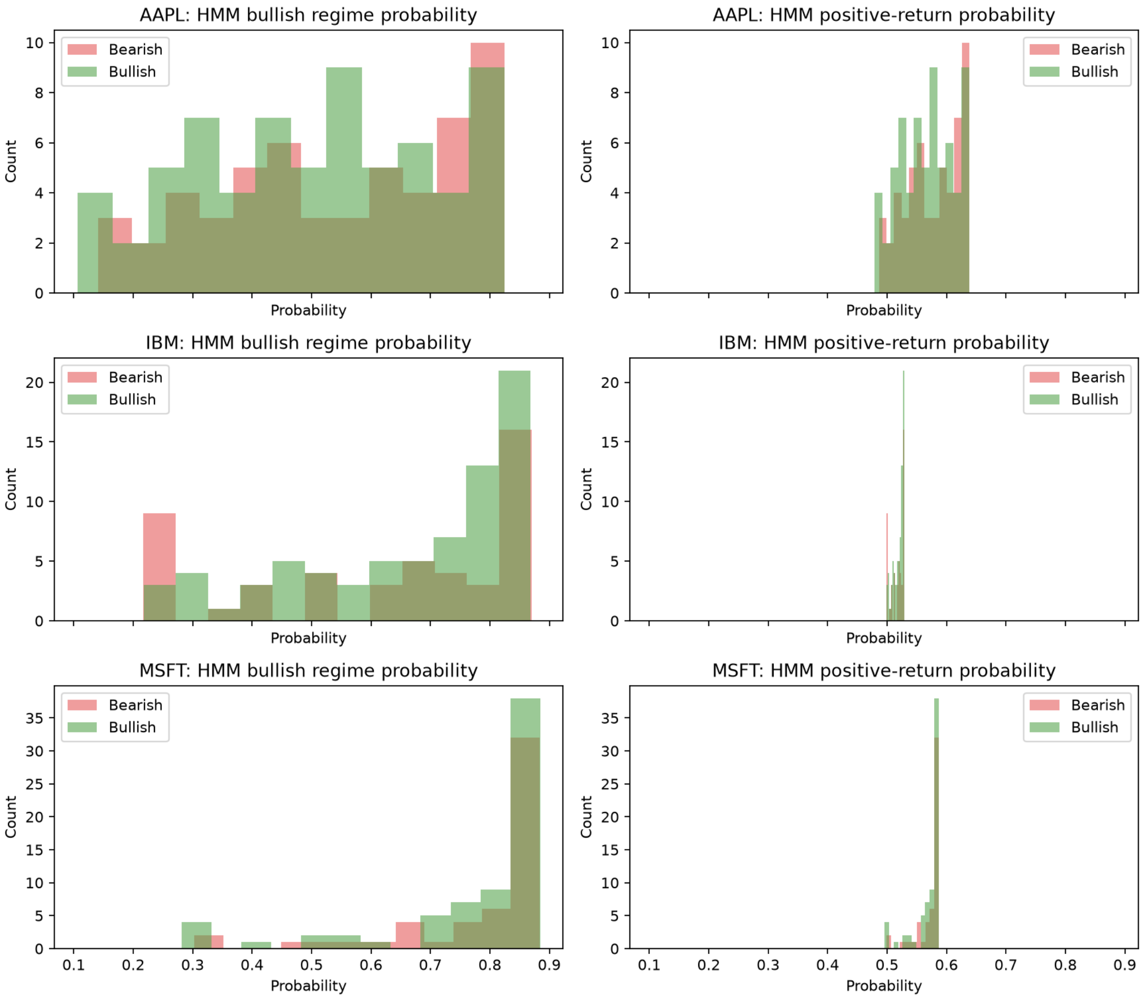


Figure 10: Test-period distributions of the HMM bullish-regime probability and HMM-derived positive-return probability, conditional on the realised next-week bearish or bullish class. The overlap between the distributions illustrates the limited directional discrimination of the HMM-only features.

B Controlled Weekly LSTM Architecture

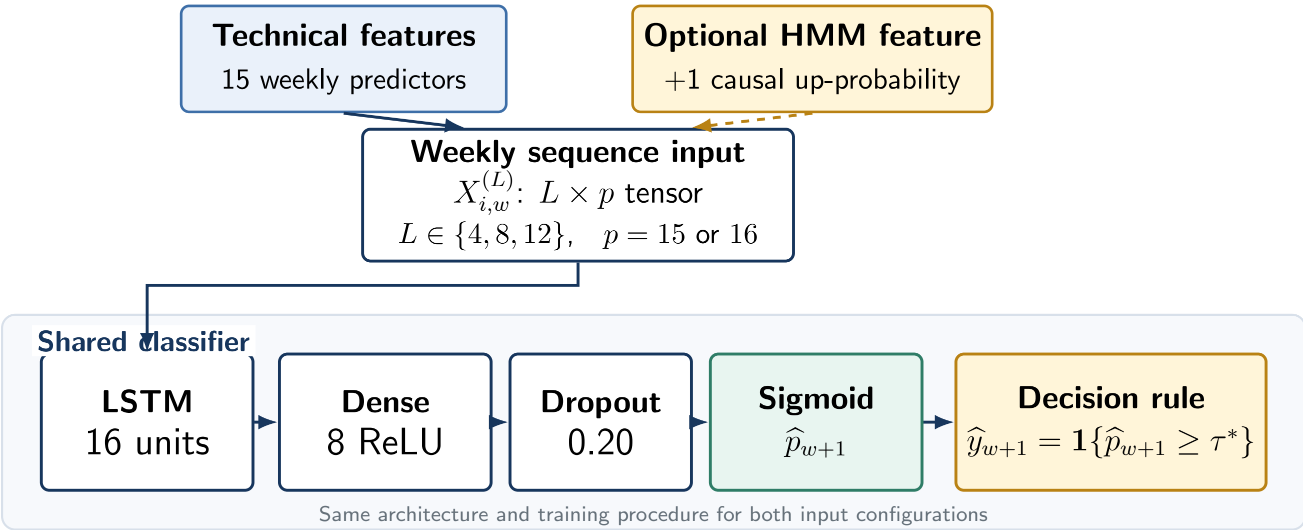


Figure 11: Architecture of the controlled weekly LSTM classifier. The technical-only model receives 15 weekly technical predictors, while the HMM-informed model adds one causal HMM-derived up-probability. Both configurations use the same 16-unit LSTM, eight-unit ReLU dense layer, dropout of 0.20, sigmoid output, and validation-selected classification rule.

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